Karan Kumar

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Education

Vellore Institute of Technology **2023 – 2027**
Bachelor of Technology - Computer Science; CGPA: 8.7 *Chennai, Tamil Nadu*

Experience

Turf (TurfLang) | Compiler, Language | Repo **Jan 2026 – Present**
Core Maintainer *Open Source*

- Core maintainer of **Turf**, a statically typed, **AOT**-compiled systems language targeting native executables via **LLVM** — structs, enums, generics, pattern matching, error unions, and a GC'd heap with checked references.
- Help ship the **turf CLI** (build/run/test/fmt/check), an **LSP** server, and a standard library covering strings, collections, files, JSON, HTTP, and sockets; each release bundles its own LLVM, linker, and runtime.
- Maintain **CI** testing every change across **4 Tier-1 platforms** (Linux x86-64/arm64, Windows x86-64, macOS arm64) gating releases; grown to **1,200+ commits**, **78K LOC**, and **v2.1.1** with a 3-person core maintainer team over 10 months.

Yes Bank **May 2026 – Jun 2026**
Artificial Intelligence Intern *Mumbai, Maharashtra*

- Engineered a low-latency **AI** conversational interface on **Azure OpenAI** with **SSE**-based token streaming and **Next.js**/**TypeScript** edge rendering for real-time enterprise workflows.
- Rewrote a legacy **Python/Streamlit** app into production-grade **Next.js**, with **JWT**-based edge auth, **RBAC** middleware, and a **SQLite**-backed governance layer for concurrency-safe bulk operations.

ThinkDeck (Aizonics) **May 2025 – July 2025**
Full Stack Developer Intern *Remote*

- Built scalable REST APIs using **Go (Gin)** and **FastAPI** powering core platform features and AI-generated responses, with **token-based auth** and rate-limiting middleware.
- Integrated **RabbitMQ** and **Redis** to handle async tasks, caching, and background job processing.

Projects

Devdoq | *Next.js, Convex, Ai SDK, Redis, Cocoindex, Zustand* | Live Link **Project — Feb 2025**

- Built an AI dev-docs platform with codebase analysis via **CocoIndex vector indexing** and **code chunking**.
- Scaled to **650+ beta testers** with talk-to-codebase, code visualization, API docs generation, and codebase mindmaps.
- Implemented **BullMQ job queues** and background workers for **multi-threaded** doc generation on large repositories.

Nuza | *Tauri, Rust, React, TypeScript, CodeMirror, Tailwind CSS* | Live Link | Repo **Open Source Project — Sep 2026**

- Built a local-first, cross-platform Markdown notes app (**Obsidian** alternative) with **Tauri**, **Rust**, and **React** — live preview, Vim keybindings, wiki-links/backlinks, and an opt-in **AI panel** over the Agent Client Protocol (no server, account, or API key).
- Engineered a **multi-threaded Rust** search engine indexing **10K notes in <200ms**, full-text/regex search over 38MB in <**90ms**; **445ms** launch (**2x faster than Obsidian**), **60% less memory**, **8MB installer** (25x smaller).
- Shipped **28 releases** via **GitHub Actions** CI/CD with signed auto-updates and **270+ Rust tests**; grown to 55+ stars, 5+ contributors, and 80+ merged PRs.

Xtoxic | *Next.js, Exa, Gemini, Vercel AI SDK* | Live Repo **Personal Project — Mar 2025**

- Built and deployed a Twitter toxicity analysis tool in **1 hour** (Exa + Gemini via Vercel AI SDK); scaled to **12,000+ users** and **50,000+ impressions** in 12 hours, open-sourced with **30+ GitHub stars**.

HTTP Server | *C, Linux, Sockets, POSIX Threads, Regex* | Live Repo | Blog **Personal Project — Feb 2025**

- Built a multi-threaded HTTP server in C handling **9,000+ RPS** at **2.8ms latency** and **99.99% uptime**, with MIME detection, URL decoding, and **60% lower memory usage**.

Technical Skills

Languages: JavaScript, TypeScript, Python, Rust, Java, C, C++, SQL, Bash, HTML/CSS
Developer Tools: VSCode, Neovim, Git, GitHub, GitHub Actions, Lazygit, Docker, Postman, tmux, ngrok, Linux
Frameworks: Node.js, Express, Hono, Next.js, React, React Native, Tauri, Cloudflare Workers, Supabase, Redis, MongoDB
Coursework: Data Structures, Operating Systems, Computer Networks, DBMS, OOP, Computer Architecture

Leadership / Extracurricular

- **TechnoVIT'26** Website Organizer, led **20 devs** building the tech fest site and event CMS (**40k+ requests**).
- Technical Core Committee Member at **IEEE ComSoc**, leading the technical team on project development.
- Led SIH 2024 team to **3rd place** among **3200+ participants**, managing deliverables and collaboration.
- Managed **WatchOut Wearables** marketing for a Shark Tank India pitch raising **1 Cr at 10 Cr valuation**.